Experiment 6: Text Generation Using OpenAI LLM API

1. Aim

To develop a Python program that connects to a Large Language Model (LLM) using the OpenAI API and generates text based on a user-provided prompt.

2. Procedure

1. Install Python and the OpenAI library.
2. Obtain an API key from the OpenAI platform.
3. Set the API key as an environment variable.
4. Import the OpenAI library into the Python program.
5. Get a text prompt from the user.
6. Send the prompt to the LLM using the OpenAI API.
7. Display the generated text as output.

3. Program (Python Code)

```
from openai import OpenAI


# Create OpenAI client
client = OpenAI()


# Get input from the user
prompt = input("Enter your prompt: ")


# Generate text using OpenAI API
response = client.responses.create(
    model="gpt-4.1-mini",
    input=prompt
)


# Display the generated text
print("\nGenerated Text:")
print(response.output_text)
```

4. Input

Enter your prompt: Write a short paragraph about Artificial Intelligence.

5. Output

Generated Text:



```
[1] !pip install -q transformers torch
from transformers import pipeline

# Load a text-generation model
print("Loading AI model... Please wait.")
generator = pipeline(
    "text-generation",
    model="distilgpt2"
)
print("Model loaded successfully!\n")

# Get input from the user
prompt = input("Enter your prompt: ")

# Generate text
result = generator(
    prompt,
    max_new_tokens=100,
    num_return_sequences=1,
    do_sample=True,
    temperature=0.7,
    pad_token_id=50256
)

# Display generated text
print("\nGenerated Text:")
print(result[0]["generated_text"])
```

```
Loading AI model... Please wait.
config.json: 100% | 762/762 [00:00<00:00, 4.21MB/s]
model.safetensors: 100% | 268M/268M [00:04<00:00, 62.1MB/s]
generation_config.json: 100% | 665/665 [00:00<00:00, 3.62MB/s]
vocab.json: 100% | 1.04M/1.04M [00:00<00:00, 6.79MB/s]
merges.txt: 100% | 456k/456k [00:00<00:00, 8.46MB/s]
tokenizer.json: 100% | 1.36M/1.36M [00:00<00:00, 7.30MB/s]
Model loaded successfully!

Enter your prompt: Write a short paragraph about Artificial Intelligence.

Generated Text:
Write a short paragraph about Artificial Intelligence. Artificial Intelligence (AI) is a technology that enables
machines to perform tasks that normally require human intelligence. It is widely used in healthcare, education,
robotics, and many other fields. AI helps automate tasks, analyze data, and improve decision-making.
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and improve decision-making.

Note: The generated text may vary depending on the model's response.

6. Result

Thus, a Python program was developed successfully to connect to a Large Language Model using the OpenAI API and generate text based on a user-provided prompt.